

PH 579: Quantum Computation and Information

Course Code	PH 579
Course Name	Quantum Computation and Information
L-T-P-C	3-0-0-3
Prerequisites	PH 513, PH 301 or equivalent
Intended for	UG/PG/I-PhD/PhD
Mutual Exclusion	NA
Approval	57th BoA (Oct 2024)

Course Contents

- **Foundations of Quantum Computing:** Introduction to Quantum Computing, Review of linear vector spaces, Review of Quantum postulates, Qubits and Bloch Sphere, Basic Quantum Gates, Quantum Circuits, Quantum No-Cloning Theorem and Teleportation, Quantum Teleportation Protocol, Quantum Dense Coding, Density Matrix I. (Lectures 1–10)
- **Advanced Quantum Concepts:** Density Matrix II, Projective Measurement, Positive Operator-Valued Measure (POVM), EPR Paradox and Bell's Inequalities I, Bell's Inequalities II, Deutsch Algorithm, Deutsch-Jozsa Algorithm, Simon's Problem, Grover's Search Algorithm I, Grover's Search Algorithm II. (Lectures 11–20)
- **Quantum Algorithms and Applications:** Grover's Search Algorithm III, Grover's Search Algorithm IV, Quantum Fourier Transform I, Quantum Fourier Transform II, Period Finding Problem, Method of Continued Fractions, Shor's Factorization Algorithm I, Shor's Factorization Algorithm II, Quantum Error Correction Codes I, Quantum Error Correction Codes II. (Lectures 21–30)
- **Quantum Information Theory and Cryptography:** Classical Information Theory, Shannon Entropy I, Shannon Entropy II, Von Neumann Entropy I, Von Neumann Entropy II, Classical Cryptography, RSA Algorithm, Quantum Cryptography: BB84 Protocol, Quantum Cryptography: B92 and Eckert Protocol. (Lectures 31–39)
- **Advanced Topics in Quantum Computing:** Practical Realization of Quantum Computers, Future of Quantum Computing, Introduction to Quantum Simulators (QISKIT, etc.). (Lectures 40–42)

Laboratory Modules

1. NA

Text Books

1. Michael A. Nielsen and Issac L. Chuang, **Quantum Computation and Information**, Cambridge, 2002)
2. Mikio Nakahara and Tetsuo Ohmi, **Quantum Computing**, CRC Press, 2008.
3. N. David Mermin, **Quantum Computer Science**, Cambridge, 2007.

References

1. NA